

When to Order Evacuation under Wildfire Forecast Uncertainty: A Sequential Bayesian Rule against Fixed Triggers in North-East Attica

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Abstract

Intent: State the decision (order a zone now or wait for a better forecast), the region and hazard, the policies compared, and the B001 outcomes of the headline claims in ARGUMENT.md order; no result before generated/ exists.

Keywords: wildfire evacuation, evacuation triggers, sequential decision under uncertainty, value of information, wildland–urban interface, traffic simulation, responder pre-positioning

1. Introduction

When a wildfire starts near the wildland–urban interface, the authority responsible for evacuation faces the same question at every moment: order the threatened neighbourhoods out now, or wait for a better picture of where the fire is going. Waiting has a value. Each observation of the front narrows the spread forecast, and a narrower forecast sends fewer people away from places the fire never reaches. Waiting also has a price. In a fast fire the time left to leave shrinks while the forecast improves, and an order given late sends residents onto roads that the fire may reach before they clear them. Evacuation failures in which people were caught on the roads have made the second risk concrete [1, 2].

Two further features make the decision hard. Orders are irreversible and they interact: an order to one zone loads the exits that other zones will need,

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so a broad early order can delay the people most at risk. And the information the decision-maker receives is itself shaped by time: the observed front is the evidence that updates the forecast, and it arrives only as the fire advances.

Research on the timing of wildfire evacuation orders has centred on trigger boundaries: perimeters around a community such that, once the fire crosses them, the time left before it arrives equals the time needed to evacuate [3]. Stochastic versions compute the boundary as a probability from variations of weather, fuel and model [4, 5]. In these studies the boundaries are computed before the event and are not revised as the front is observed. Sequential rules that weigh ordering now against waiting for a better forecast exist for hurricanes [6] and for a generic hazard forecast in probability [7], but not, to our knowledge, for wildfire zones coupled by congestion and informed by the observed front (Section 2).

This paper asks whether such a rule is worth having, where its value comes from, and how that value depends on the fire. We study wildfire in north-east Attica, Greece, the interface between Penteli, Dionysos, Marathon, Nea Makri and Rafina, where the Mati fire of 2018 and the fire of August 2024 give two anchor events (Section 3.1) and the regional fire record gives many more ignitions. Fire scenarios are drawn from that record, the forecast of each scenario is updated with its observed front, and every policy is evaluated in a mesoscopic traffic simulation of the region’s road network. The cost of a policy weighs the people exposed to the fire against the people ordered out of zones the fire never reaches, at a cost ratio fixed before any run. Every endpoint, margin and decision rule was fixed in a pre-registered protocol before any scenario of the region was drawn, and every endpoint is read on held-out test scenarios only.

The contributions are stated as the claims the experiment tests; each is a prediction with a pre-stated rule for support, weakening, refutation or inconclusiveness (Section 6).

1. **A sequential rule against tuned fixed triggers (C1), and where its gain comes from (C5).** A sequential Bayesian rule, which orders a zone when the expected cost of ordering now is below the expected cost of waiting one more decision epoch for a better forecast, congestion included, is predicted to have a lower expected decision cost on held-out scenarios than the best fixed trigger, whether that trigger is a distance or time buffer or a fixed probability threshold, each tuned on training scenarios at the same cost ratio. A value-of-information ladder then

splits the gain over the trigger boundary computed once into the value of updating that boundary with the observed front and the value of planning with the cost of waiting and congestion, and predicts that updating is worth more than planning.

2. **Lead time against forecast skill (C2).** Earlier detection is predicted to lower the cost more than halving the forecast error in fast scenarios, and the reverse in slow ones; and the advantage of lead over skill is predicted to fall as the scenario’s evacuation safety margin, the fire’s arrival at the first populated zone over that zone’s clearance time, grows.
3. **Forecast error and order breadth (C3b).** At equal false alarms, the advantage of staged over simultaneous area-wide evacuation is predicted to be larger when the forecast error is halved.

Two further claims are tested and reported with lower weight. That staged evacuation leaves fewer people exposed than simultaneous evacuation at equal false alarms (C3) is a replication, under forecast error, of a direction already reported for Mati [8]. That moving responder units to the staging points of highest posterior risk lowers the cost more than fixed stations (C4) is secondary, because its effect is set by the responder effect the model assumes (Section 5.4). A learned policy is reported as a comparator without a claim.

The paper does not estimate fatalities: exposure, the number of people present in a zone or on a road link when the fire arrives there, is a proxy for harm. It makes no claim beyond the region and the hazard studied, and it passes no judgement on the decisions taken during the historical events, which serve for validation only.

Section 2 places the work. Section 3 describes the region, the scenarios, the population and the traffic model; Section 4 the forecast and the decision cost; Section 5 the policies; Section 6 the pre-registered endpoints, decision rules and checks.

2. Related work

Trigger boundaries for wildfire evacuation. A trigger boundary is a perimeter around a community at which the approaching fire is as far away in time as the evacuation takes; once the fire crosses it, the order must be given. Mitchell et al. [3] design such triggers by combining spread and evacuation times in the PERIL model. Kalogeropoulos et al. [4] make them stochastic:

the k-PERIL tool computes, from historical variations of wind, weather and vegetation, the probability that a landscape cell lies inside the boundary. Kalogeropoulos and Rein [5] obtain probabilistic triggers from an ensemble of spread models, find that under rapid spread the triggers of their case study lie beyond practical detection distances, and argue that shelter-in-place must be considered beside evacuation. Kalogeropoulos et al. [1] define an evacuation safety factor from the probabilistic race between fire and evacuation and find that, at Mati in 2018, it was well below one from the moment the fire was detected. In these studies the boundary is computed before the event. The boundary is not revised as the front is observed, false alarms are not priced, and the orders of one community do not change the evacuation time of another. We take the stochastic boundary as one of the fixed triggers the sequential rule must beat, computed once from the prior forecast as in k-PERIL, and we add the same boundary recomputed at each epoch from the updated forecast, so that the value of updating can be separated from the value of planning (Section 5). The Mati finding of Kalogeropoulos et al. [1] is an independent check for our first anchor event, and their safety factor motivates the evacuation safety margin by which we index the crossover of lead time and forecast skill, without our margin claiming to equal it. That lead time dominates in fast fires is close to what trigger-boundary analyses already imply; our contribution there is the slow case and the crossover between the two.

Staged and simultaneous evacuation. That staging orders relieves the congestion a simultaneous order creates is an established result for wildfire evacuation. Hosseini et al. [9] optimize a staged schedule under congestion and fire threat, and Maha Gamage et al. [10] build a staged model for interface communities with transient populations. For Mati itself, Siam et al. [8] report, with an agent-based model coupling fire, population response, the transport network and shelters, that staged evacuation improves life safety. These studies work with a known fire. Our comparison of staged and simultaneous evacuation is therefore a replication, under forecast error and at equal false alarms, of a published direction. What a known fire cannot show is how the advantage of staging changes with the quality of the forecast that ranks the zones; that interaction is the new question we test.

Ordering now or waiting for a better forecast. The trade-off between acting early on an uncertain forecast and waiting for a better one has been studied

for other hazards. Yi et al. [6] optimize the issuance of evacuation orders as a hurricane evolves, with evacuee behaviour and traffic in the model. Georgakakos [7] analyse, for a generic life-threatening hazard forecast in probability in real time, the decision to issue an evacuation statement now or delay it, accounting for the forecast sequence, the public response, the completion time of the evacuation and the risk aversion of the public and of the agency. The concept of a rule that orders now or waits is thus not new. What these works do not treat is the wildfire case: a spread forecast updated with the observed front on a time scale of minutes, several zones whose orders compete for the same exits, and a decomposition of the rule’s gain into the value of updating and the value of planning. Those three elements are what our sequential rule and its value-of-information ladder add.

Coupled fire and evacuation simulation. Evaluating an evacuation policy requires the fire, the people and the traffic in one model. Open platforms couple these layers for the wildland–urban interface, among them WUI-NITY [11], and recent agent-based frameworks add multimodal transport and dynamic routing [12]. Because such models are costly to run many times, learned surrogates of the traffic layer have been proposed for probabilistic evacuation planning [13]. We use a simulator of this kind as the evaluator of every policy and do not contribute to simulation itself; the planning model inside the sequential rule is a separate, cheaper approximation, and no endpoint is computed from it (Section 5).

Pre-positioning of responders. Where to place response resources before and during a fire is mostly studied for suppression. Alvelos et al. [14] formulate a mixed-integer model for dispatching, positioning and routing firefighting crews and aircraft in the initial attack, with spread modelled as shortest paths in a network, and Avcı et al. [15] treat suppression with several types of resource. In those models the resources act on the fire. The responder units of this paper act on the evacuation instead: a unit at a staging point raises the discharge of the road it staffs, and in one arm the units are moved to the points of highest posterior risk as the forecast is updated. How much a staffed point raises discharge is an assumption of our model, not an estimate from data (Section 5.4). The comparison of moving and fixed units is therefore secondary, and its result is read as conditional on that assumption.

Behaviour during wildfire evacuation. Warning response, preparation and departure are reviewed by Kuligowski [16], and departure timing has been

estimated from large-scale GPS data [17]. For the study region, Siam et al. [18] surveyed Mati households about their expected evacuation logistics and time components, which calibrates our preparation times, and Zikeloglou et al. [19] report residents of north-east Attica who stayed to protect their property in the fires of 2021, a reason to treat compliance as uncertain. In a simulation of a fast-moving fire, Grajdura et al. [2] find that delays in awareness raise entrapment, the behavioural face of lost lead time.

3. Region, hazard scenarios and population

3.1. Study region and anchor events

The study region is the wildland–urban interface of north-east Attica, Greece, covered by the municipalities of Penteli, Dionysos, Marathon and Rafina–Pikermi ([[polygon-definition](#)]). The region is split into evacuation zones, which are settlement or neighbourhood polygons ([[n-zones](#)]), and safe destinations are fixed points outside the region on its main exits, set before any run.

Two historical fires in the region serve as anchor events, for validation only; neither enters the training or the test set. In the Mati fire of 2018, fast-moving flames led to a dire evacuation with many fatalities, and the evacuation safety factor of Kalogeropoulos et al. was well below one from the moment the fire was detected [1]. The evacuation of Mati has been modelled with an agent-based model coupling the fire, the population’s response and the transport network [8], and households of the area were surveyed afterwards about their expected evacuation logistics and time components [18]. No evacuation order was issued at Mati ([[source-mati-no-order](#)]). The second anchor is the wildfire of August 2024 in north-east Attica [20], which started near Varnavas ([[source-varnavas-ignition](#)]); during it the authorities ordered the evacuation of several settlements in sequence through cell-broadcast messages of the European emergency number ([[source-varnavas-cell-broadcast](#)]). The first event marks one end of the decision this paper studies, no order at all, and the second the other, orders issued settlement by settlement. Both events were mapped by the Copernicus Emergency Management Service (activations EMSR300 and EMSR746; [[cite-emsr-datasets](#)]), whose burnt-area delineations are the validation targets below. The region also burned in 2021, when some residents of north-east Attica stayed to protect their property rather than leave [19]; those fires are not anchors.

3.2. Fire model and scenario generator

Fire spread is simulated with `[fire-model]` on fuels from the European Forest Fire Information System fuel map or CORINE Land Cover mapped to the standard fuel models of Scott and Burgan, on the Copernicus GLO-30 digital elevation model, with hourly ERA5 reanalysis weather. Dead-fuel moisture follows the hourly weather through an equilibrium-moisture relation, and live-fuel moisture is set to a fully cured state, the same for every scenario (`[fuel-moisture-rule]`). A scenario is the field $T(x)$ of fire-arrival times over the region produced by one fire run, from ignition to a horizon of `[horizon]`.

Anchor validation. Before any policy is run, each anchor is simulated from its documented ignition point and weather, and checked against its rapid-mapping delineation (Sørensen index at least `[sorensen-min]`) and against documented arrival times at named places (within `[arrival-tolerance]`). If an anchor fails, the fire model is recalibrated on the anchors only, through the central spread parameters and never through the forecast-error scale of Section 4.1; the failure and the recalibration are reported.

Scenario generator. Ignition points are drawn from a kernel density of the fires recorded by the European Forest Fire Information System in and around the region over `[effis-period]`, and weather from the summer days of that period in the top `[fwi-quantile]` of the Canadian Fire Weather Index at the ERA5 cell nearest the region's centroid, with the moisture codes run continuously from a spring start (`[fwi-rule]`). Scenarios are drawn in sequence from a registered seed and all drawn scenarios are kept. The first `[n-train]` form the training set. The test set is drawn next, until `[n-test]` scenarios reach at least one populated zone, with at most `[max-draws]` draws in all; scenarios that reach no populated zone and are drawn meanwhile stay in the test set, where they can only generate false alarms. The test set is enlarged to `[n-test-enlarged]` reaching scenarios (at most `[max-draws-enlarged]` draws) under the sample-size rule of Section 6.2. Every tuning and every sample-size decision uses the training set only.

3.3. Population and behaviour

Residents per zone come from the Global Human Settlement Layer population grid or the 2021 Greek census. Persons per vehicle is the census mean household size over the mean number of vehicles that Mati households expect

to take [18], bounded below by one ([persons-per-vehicle]). Zones flagged as second-home areas carry [second-home-factor] times their resident population in a secondary arm.

Ordered residents comply with probability c ([compliance-primary]); sensitivity [compliance-sensitivity] and leave after a preparation time drawn from a lognormal distribution fitted to the expected preparation times reported by Siam et al. [18] ([prep-time-fit]). Everyone, ordered or not, leaves spontaneously when the front comes within [spontaneous-distance] of their zone, after the same preparation time. Warning receipt is not modelled separately. Because the survey measures expected rather than realized times, a secondary arm multiplies preparation times by [slow-prep-factor]. Behaviour is random: each (scenario, policy point) is run with [n-seeds] behavioural seeds, common to all policies, and a scenario’s value is the mean over its seeds.

3.4. Traffic

Evacuation traffic is simulated with Eclipse SUMO¹ in its mesoscopic mode ([sumo-version]), on a network built from an OpenStreetMap snapshot ([osm-snapshot]). Policies act between decision epochs through the simulator’s control interface. Road links inside burning cells close, and vehicles on them reroute. SUMO is the evaluator of record: every endpoint, every tuning of a fixed trigger and every check in this paper is computed from SUMO runs, never from the planning surrogate of Section 5.

The mesoscopic model gives vehicles leaving a jammed segment a longer headway than vehicles leaving a free one, so a queue that spills back from a bottleneck discharges more slowly than a free-flowing one: a capacity drop is built into the evaluator. Before any run on the region, a bottleneck benchmark with vehicles inserted upstream at a rate above capacity separates this property of the model from an insertion artefact, and the discharge as a function of queue length is reported ([bottleneck-result]). Because this drop is what makes a broad order costly, a secondary arm removes it (jammed and free headways set equal) and is reported beside the staged-versus-simultaneous comparison (Section 6.1).

Road exposure is counted per edge of each vehicle’s route when the fire reaches that edge, which is slightly conservative.

¹<https://eclipse.dev/sumo/>

4. Decision problem

The region is partitioned into zones $z \in \mathcal{Z}$ with known population. A hazard scenario is a fire-arrival-time field $T(x)$ over landscape cells x ; zone z is *reached* if the fire arrives in it before the horizon, and $T(z)$ denotes its arrival time there. The decision-maker acts at epochs $t = d_0, d_0 + \Delta t, \dots$ up to the horizon, where d_0 is the delay between ignition and the first forecast (the detection delay, or *lead*) and $\Delta t = \text{[epoch-interval]}$.

4.1. Forecast and its updating

Each scenario’s truth is a fire run with spread parameters θ^* : a rate-of-spread multiplier, a wind-direction offset and a wind-speed multiplier. The forecaster does not know θ^* . Its prior over θ is normal, centred on $\theta^* + e$ with $e \sim N(0, s^2 \Sigma_0)$ and with that same covariance $s^2 \Sigma_0$, so that the prior is calibrated; $\Sigma_0 = \text{diag}(\sigma_0^2)$ is the nominal error scale and s the skill setting. The rate-of-spread component of σ_0 is taken from a multi-dataset evaluation of spread-model error and the wind components from a study of large Mediterranean fires, each converted to one standard deviation by a rule fixed before training ([\[sigma0-values\]](#); [\[cite-sigma0-sources\]](#)).

The forecast is an ensemble. For each scenario a fixed bank of member runs m covers a grid of θ around the prior centre ([\[member-grid\]](#)), each member with arrival field $T_m(x)$ and prior weight w_m^{prior} . At epoch t the members are reweighted by their agreement with the observed front, the burnt area of the truth at t minus an observation latency of [\[obs-latency\]](#), with a likelihood tempered by a factor κ . The posterior weights w_m^t give, for each zone, a posterior distribution of its arrival time. The factor κ is set on the training set only, as the value in [\[kappa-grid\]](#) whose central posterior interval ([\[kappa-coverage-level\]](#)) of zone arrival covers the true arrival at the rate closest to its nominal level; calibration, not sharpness, is what the prior requires. The effective sample size of the weights is reported per epoch.

Skill, lead and misspecification. Three skill settings are used: nominal error ($s = s_{\text{nom}}$), error halved ($s = s_{\text{nom}}/2$), and no error ($s = 0$, the oracle forecast, used only in checks). Three lead settings are used: the nominal first epoch $d_0 = \text{[d0-nominal]}$ after ignition, a lead gain of [\[lead-gain\]](#) and an intermediate gain of [\[lead-gain-mid\]](#). In the *misspecification* arm the forecaster states an error spread of [\[misspec-spread-factor\]](#) times its true error and its wind direction is biased by [\[misspec-wind-bias\]](#); its prior is then overconfident and off-centre.

4.2. Actions, outcomes and decision cost

At each epoch the decision-maker, for each zone not yet ordered, either waits or orders its evacuation; an order is irreversible. In the responder arms it also chooses, for each of K units, whether to stay or to move to a staging point (Section 5.4).

Three outcomes are recorded per run. *Exposure* E is the number of people present in a zone cell, or on a road link, when the fire arrives there; road exposure, the entrapment of Mati, is also reported on its own. *False-alarm persons* A is the number of people ordered out of a zone that the fire does not reach before the horizon. *Unit-hours* is the time responder units spend deployed. The decision cost is

$$J = E + r A, \quad (1)$$

where the cost ratio r states how many people evacuated in vain weigh as much as one person exposed. It is fixed before any result: primary $r = [\text{r-primary}]$, sensitivity $r \in [\text{r-sensitivity}]$. The full exposure–false-alarm frontiers are also reported, so that a reader can apply another ratio. Unit-hours are reported but not priced: they are equal across the arms that compare responder placements, and responders are held at static stations in every other comparison.

Relative differences between policies are ratios of totals over the test set, $\Delta = (J_{\text{ref}} - J_{\text{new}})/J_{\text{ref}}$.

5. Policies compared

All policies see the same forecast at each epoch and act on the same scenarios, behavioural seeds and road network. The parameter of each fixed family is tuned on the training set at each value of r , by the training J in SUMO, over a grid fixed before any run. The fixed families differ in what they read from the forecast and in the form of the rule; Table 1 shows how three of them, with the sequential rule, form a ladder that separates the value of updating from the value of planning.

Static clearance time. Each fixed family compares the fire’s arrival with the zone’s static clearance time c_z , the time needed to clear the zone at its own load. It is computed with the same calibrated exit discharge that the sequential rule plans with (Section 5.2), so that the comparison is not won by a better clearance model on one side.

5.1. Fixed triggers

F1, buffer trigger. Following trigger practice [3], zone z is ordered when its forecast median arrival time is within $c_z + b$, with the buffer b tuned over **[grid-b]**.

F2, fixed probability threshold. Zone z is ordered at the first epoch at which the posterior probability that the fire reaches it within c_z is at least q , with q tuned over **[grid-q]**. This is the cost-loss form of a forecast-based order, re-read at every epoch from the updated forecast.

F2b, stochastic trigger boundary computed once. This family implements the probabilistic trigger boundary of k-PERIL [4]. For zone z and member m , the member’s boundary is the set of cells its front crosses within c_z before reaching z ,

$$B_m(z) = \{x : 0 \leq T_m(z) - T_m(x) \leq c_z\},$$

empty if the member never reaches z . The boundary probability is $p_z(x) = \sum_m w_m^{\text{prior}} \mathbf{1}[x \in B_m(z)]$, computed once at the first epoch from the prior weights and never revised. Zone z is ordered at the first epoch at which the observed burnt area contains a cell with $p_z(x) \geq q$, with the same observation latency as the forecast, no buffer, and q tuned over the grid of F2. One deviation from k-PERIL is deliberate: k-PERIL’s reverse spread covers every cell from which a fire could arrive in time, whatever its ignition, whereas here only cells on the members’ paths from the known ignition count. The boundary is therefore narrower, and a true fire that passes where no member went does not trigger it; this weakens F2b, mostly under misspecification, and is stated beside any result in which F2b is the best fixed family.

F2b-u, stochastic trigger boundary recomputed. F2b with the prior weights replaced, at each epoch, by the posterior weights: $p_z^t(x) = \sum_m w_m^t \mathbf{1}[x \in B_m(z)]$. The trigger, latency, buffer and grid are those of F2b, and q is tuned in the same way. F2b-u serves only the value-of-information ladder; it is not one of the fixed triggers that the sequential rule is compared with in the primary endpoint (Section 6.1).

5.2. Sequential Bayesian rule

F3. At each epoch, zones are visited in order of risk, and zone z is ordered now if the expected J of ordering it now is below the expected J of waiting one epoch and then acting on the updated forecast. The expectation is a

one-step look-ahead over the preposterior: each member, weighted by its posterior weight, is treated as a possible truth, and the forecast it would produce one epoch later is computed. Orders already given, and the order under consideration, load the exits, so the congestion an order causes enters its cost. The rule has no parameter besides r .

Planning surrogate. Planning with SUMO inside the look-ahead would need one traffic run per zone, epoch and member, far beyond the budget. F3 therefore plans with a fast exit-queue surrogate: a load-dependent discharge at each exit, an internal delay term for congestion before the exit, and road exposure counted per edge of the static route ([[surrogate-form](#)]). The surrogate is calibrated once on SUMO runs of training scenarios only, then frozen before any test run. Its calibration targets, fixed before fitting and checked by leaving one training scenario out, are: zone clearance time within a median absolute relative error of [[T1-clearance-tol](#)]; pooled road exposure within [[T2-road-tol](#)] of SUMO’s, without the same sign of error in more than [[T2-sign-share](#)] of scenarios; and a median Kendall rank correlation of at least [[T3-kendall-min](#)] between the surrogate’s and SUMO’s ranking of policy points by J . If a target is missed, the best-fitting surrogate is still frozen and used, and the miss is reported beside the endpoints it can affect. A residual planning error handicaps F3, since the fixed families are tuned in SUMO and absorb its behaviour through their parameter while F3 has none; it biases the primary comparison against the sequential rule.

5.3. Simultaneous evacuation, learned comparator and references

S, simultaneous area-wide evacuation. At the first epoch at which the probability that any populated zone is reached is at least [[S-activation](#)], every zone whose reach probability is at least q_ℓ is ordered at once, with the breadth q_ℓ ranging over [[grid-q-ell](#)]. S is not tuned to a single point: its whole exposure–false-alarm frontier over q_ℓ is used for the comparison at equal false alarms.

L, learned comparator. A policy trained by proximal policy optimization on zone features (arrival quantiles, population, exit queues), on the training scenarios, in the planning surrogate, and evaluated in SUMO; its training budget is capped at [[L-budget](#)]. It is reported beside the others without a claim.

Table 1: The value-of-information ladder. Each step changes one element of the rule, all on the same member bank.

Policy	Information	Rule form	Step from the row above
F2b	prior, computed once	front crosses a boundary	—
F2b-u	posterior, each epoch	front crosses a boundary	updating (v_{upd})
F2	posterior, each epoch	zone probability threshold	rule form (v_{form} , reported)
F3	posterior, each epoch	look-ahead with congestion	planning (v_{plan})
oracle	true arrival field	latest clearing order	perfect information left (v_{pi})

References. The *no-order* policy leaves only spontaneous departures. The *oracle* knows the true arrival field and orders each reached zone at the latest time that clears it, and also any unreached zone whose evacuation route burns when ordering it lowers the surrogate’s J ; it plans with the calibrated surrogate, so it is an oracle of the hazard, not of traffic.

5.4. Responder pre-positioning

$K = [\text{K-units}]$ responder units act on throughput only: while a unit is at a staging point, vehicles on that staffed edge discharge faster by a calibrated fraction ($[\text{responder-gain}]$; sensitivity $[\text{responder-gain-sensitivity}]$). The fraction is realized in the mesoscopic model through the reaction time of vehicles on the staffed edge, restricted to the range where discharge is monotone in it and calibrated on the region’s staging edges per lane-count class at two loads ($[\text{responder-calibration}]$). This is a modelling choice: nothing is claimed about how traffic control achieves the gain, and door-to-door warning is not modelled, which understates what responders can do. In the *static* arm, units stay at stations chosen to cover the historical ignition density. In the *dynamic* arm, at each epoch, units move to the staging points with the highest posterior expected exposure, read from the surrogate’s risk map. Both arms use the sequential rule and the same unit-hours; a no-unit arm is reported as reference.

6. Experimental design and evaluation

Every endpoint, margin, grid and decision rule below was fixed in a pre-registered protocol before any scenario of the region was drawn, and may not be changed after the first test run. Endpoints are computed on the test scenarios only, with every policy at its training-tuned parameter. The outcome of each claim is one of *supports*, *weakens*, *refutes* or *inconclusive at this power*, read by the rule stated for its endpoint; a claim is never reworded to fit a result, and negative outcomes are reported beside positive ones.

Primary arm and secondary arms. The primary arm has $r = \text{[r-primary]}$, nominal skill and lead, primary compliance, a well-specified forecast and static responder stations. Reported beside it, and never changing an outcome except where a rule below says so, are the misspecification arm, the two sensitivity values of r , the compliance sensitivity, the second-home arm, the slow-preparation arm and the responder-gain sensitivities.

Inference. The scenario is the unit. Confidence bounds come from a paired bootstrap over test scenarios ([n-bootstrap] resamples) in which all policies are resampled together, each scenario's value being the mean over its behavioural seeds, so that behavioural noise is inside the intervals. A bound is the one-sided bound at the level of its endpoint; a "lower bound above zero" is equivalent to a one-sided bootstrap p -value below that level.

6.1. Endpoints and decision rules

Multiplicity. Four primary endpoints, E1 to E4, one per claim C1 to C4, form one family: the condition "lower bound above zero" in each of their rules uses the one-sided bootstrap p -value Holm-adjusted across E1–E4 at family level [alpha-family] . Three secondary endpoints, E5, E2c and E3b, are each read at one-sided level [alpha-secondary] , outside the family; they never change the outcome of E1–E4.

E1, sequential rule against the best fixed trigger (C1). The reference is the fixed family among F1, F2 and F2b with the lowest training J , at its tuned parameter; $\Delta_1 = (J_{\text{ref}} - J_{\text{F3}})/J_{\text{ref}}$ in the primary arm, with practical margin $\delta_1 = \text{[delta1]}$. *Supports:* lower bound of Δ_1 above zero and point estimate at least δ_1 . *Weakens:* lower bound above zero but point estimate below δ_1 ; or support in the primary arm but a point estimate of Δ_1 at most zero in the misspecification arm or at either sensitivity value of r . *Refutes:* upper bound

of Δ_1 at most zero. *Inconclusive*: otherwise. If F2b-u has a lower training J than the reference, this is stated beside E1; the reference set is not changed.

E2, lead time against forecast skill by speed class (C2). A test scenario is *fast* if its true fire reaches the first populated zone within **[fast-class-limit]** of ignition, and *slow* otherwise. For the sequential rule, in each class, $G_{\text{lead}} = J(\text{nominal}) - J(\text{lead gain})$, $G_{\text{skill}} = J(\text{nominal}) - J(\text{error halved})$, and $D = (G_{\text{lead}} - G_{\text{skill}}) / J(\text{nominal})$. The prediction is $D_{\text{fast}} > 0$ and $D_{\text{slow}} < 0$. *Supports*: both signs as predicted, each with its bound excluding zero. *Weakens*: one class as predicted with its bound excluding zero, the other's interval containing zero. *Refutes*: either class with its bound excluding zero on the wrong side. *Inconclusive*: neither interval excludes zero; a class with fewer than **[min-class-size]** test scenarios counts as an interval containing zero. The same quantities for F2 are reported as a check on planning error: a sign disagreement between F3 and F2 is stated.

E3, staged against simultaneous evacuation (C3, replication). Staged evacuation is F2, zone by zone; simultaneous is S. The false-alarm levels are the pooled test A that F2 reaches at $q \in \text{[E3-q-levels]}$; S's exposure at each level is read by monotone interpolation of its pooled test frontier (A, E) . With $\Delta_3 = (E_S - E_{F2}) / E_S$ at the middle level and margin $\delta_3 = \text{[delta3]}$: *Supports*: lower bound above zero and point estimate at least δ_3 . *Weakens*: lower bound above zero but point estimate below δ_3 ; or support at the middle level but a point estimate at most zero at either other level. *Refutes*: upper bound at most zero. *Inconclusive*: otherwise. Reported without an outcome: the share of test scenarios in which S's exposure rises as its breadth grows over part of its grid; Δ_3 in the slow-preparation arm; and Δ_3 with the capacity drop removed. If E3 supports C3 but the point estimate without the capacity drop is at most zero, the support is stated to rest on the capacity drop; the outcome is not changed, since real bottlenecks do show a drop.

E4, dynamic against static pre-positioning (C4, secondary). Both arms use the sequential rule and K units for the same duration; $\Delta_4 = (J_{\text{static}} - J_{\text{dynamic}}) / J_{\text{static}}$, margin $\delta_4 = \text{[delta4]}$. E4 stays in the family of primary endpoints as registered, but C4 is reported as a secondary claim: its effect is set by the responder gain the model assumes (Section 5.4). A test scenario is *responder-sensitive* if static units change its J , relative to the no-unit reference, by more than the behavioural noise band defined in Section 6.3. Scenarios are not filtered on this property, which would select on

the mechanism of the outcome; instead, a reading rule applies. If fewer than **[min-responder-sensitive]** test scenarios are responder-sensitive, E4 is *inconclusive at this power*, whatever its bounds: the scenarios do not load the staging points. The threshold is the smallest number at which an effect confined to the sensitive scenarios, however large, can reach the power required of every endpoint at the worst-case Holm level; it does not depend on the size of the test set. Otherwise E4 is read on all test scenarios. *Supports*: lower bound above zero and point estimate at least δ_4 . *Weakens*: lower bound above zero, point estimate below δ_4 . *Refutes*: upper bound at most zero. *Inconclusive*: otherwise. With fewer than **[low-power-responder-sensitive]** responder-sensitive scenarios the outcome is flagged as low power, without changing it. Whatever the reading, the number of responder-sensitive scenarios is reported beside E4, with the exact one-sided sign-flip p -value over those scenarios and the same count computed with the dynamic arm in place of the static one. E4 is also reported in the second-home arm, where the load is higher.

E5, value-of-information ladder (C5, secondary). In the primary arm, with totals over the test set and F2b as the base (Table 1):

$$\begin{aligned} v_{\text{upd}} &= \frac{J(\text{F2b}) - J(\text{F2b-u})}{J(\text{F2b})}, & v_{\text{form}} &= \frac{J(\text{F2b-u}) - J(\text{F2})}{J(\text{F2b})}, \\ v_{\text{plan}} &= \frac{J(\text{F2}) - J(\text{F3})}{J(\text{F2b})}, & v_{\text{pi}} &= \frac{J(\text{F3}) - J(\text{oracle})}{J(\text{F2b})}. \end{aligned}$$

The prediction is $v_{\text{upd}} > 0$, $v_{\text{plan}} > 0$ and $v_{\text{upd}} > v_{\text{plan}}$; v_{form} and v_{pi} are reported without a prediction. *Supports*: lower bounds of v_{upd} , v_{plan} and $v_{\text{upd}} - v_{\text{plan}}$ all above zero. *Weakens*: both signs supported but the interval of $v_{\text{upd}} - v_{\text{plan}}$ contains zero; or exactly one of v_{upd} , v_{plan} has its lower bound above zero while the other's interval contains zero. *Refutes*: the upper bound of v_{upd} or of v_{plan} at most zero, or the upper bound of $v_{\text{upd}} - v_{\text{plan}}$ below zero. *Inconclusive*: otherwise. The ladder is also reported in the misspecification arm and at the sensitivity values of r . If F2b is not the best fixed family, the base of E5 and the reference of E1 differ, and both are stated.

E2c, crossover of lead and skill (C2, secondary). For each test scenario i whose fire reaches a populated zone, the evacuation safety margin is $m_i = T_{\text{first}}/c_{\text{first}}$, the true arrival time from ignition at the first populated zone

reached over that zone's static clearance time. With

$$D_i = \frac{G_{\text{lead},i} - G_{\text{skill},i}}{\max(J_i(\text{nominal}), 1)}$$

for the sequential rule (the floor of one person avoids a division by zero), the prediction is a negative Spearman correlation $\rho(D, m)$. *Supports*: upper bound of ρ below zero. *Refutes*: lower bound above zero. *Inconclusive*: otherwise, or fewer than [min-reaching-E2c] reaching test scenarios. Reported without an outcome, by tercile of m : the exchange rate, i.e. the lead gain, interpolated between the lead settings, at which G_{lead} equals G_{skill} , or "beyond the largest gain" when it is not reached. E2 remains the primary reading of C2.

E3b, forecast error and order breadth (C3b, secondary). The two frontiers of E3 are re-run on the test set with the error halved, and $\Delta_3(s)$ is computed as in E3 at the middle false-alarm level of that arm. The prediction is $\Delta_3(s_{\text{nom}}/2) > \Delta_3(s_{\text{nom}})$. *Supports*: lower bound of $\Delta_3(s_{\text{nom}}/2) - \Delta_3(s_{\text{nom}})$ above zero. *Refutes*: upper bound below zero. *Inconclusive*: otherwise. Δ_3 in the misspecification arm is reported without an outcome. The oracle forecast is not used here, because without forecast error neither family produces false alarms and the comparison at equal false alarms degenerates. A mechanism is offered for C3b, that staging relies on the forecast's ranking of zones while a broad order hedges against misranking, but E3b measures only the interaction, and the mechanism is stated as an explanation, not a finding.

Arms not run. If the compute budget forces an arm to be dropped, by a pre-stated order, the endpoint that needs it is reported as *not run*, never as inconclusive.

6.2. Sample size and behavioural seeds

With [n-test] reaching test scenarios and the worst-case Holm level, a paired design has [power-target] power for a standardized paired effect of [power-dz]. Because scenario totals are heavy-tailed, a sample-size rule is applied on the training set before any test run: if the projected half-width of the [halfwidth-level] interval of Δ_1 at [n-test] test scenarios exceeds [halfwidth-max], the test set is enlarged as in Section 3.2. A seed rule is applied in the same way: the variance of per-scenario paired differences for E1 is split into a between-scenario part and a behavioural part; if the behavioural part exceeds

[behav-share-max] of the total, the number of seeds is raised to [n-seeds-raised] for every arm, within the compute cap. Both decisions and their inputs are logged before any test run. Every endpoint is inconclusive at this power if fewer than [min-reaching-all] test scenarios reach a populated zone after the draw cap.

6.3. Validation and checks

Behavioural noise band. For a paired difference on one scenario, the band is

$$\max\left(\text{persons of [noise-band-households] households, [noise-band-sd-mult] SD}_{\text{seed}}/\sqrt{n_{\text{seeds}}}\right),$$

where SD_{seed} is the seed-to-seed standard deviation of that difference pooled over training scenarios.

Checks that are not evidence. A failure of any of these is a bug or a calibration problem and is reported before any endpoint is read: the oracle has the lowest J on every scenario, and on the pooled training total, up to the noise band; with no forecast error the sequential rule’s training J is within the noise band of the oracle’s; the no-order policy has no false alarms. The anchor validation (Section 3.2), the bottleneck benchmark (Section 3.4), the calibration of the responder gain (Section 5.4) and the surrogate’s calibration targets (Section 5.2) are reported with them.

Known directions of bias. Stated before any run, for each claim. For C1, a residual planning error of the surrogate handicaps F3, and the narrower boundary of F2b favours it; the second is bounded because the reference is the best fixed family. For C2, planning error is common to the nominal, lead and skill arms and partly cancels, but in congested fast scenarios it makes F3 under-use extra lead, against the prediction. For C3, the capacity drop of the mesoscopic model, shorter preparation times and fewer persons per vehicle all raise the peak load of a broad order and favour the claim; the arms without capacity drop and with slow preparation are the checks. For C4, planning error misplaces dynamic units and door-to-door warning is not modelled; both work against the claim.

Anchor audits. After the anchors pass and every policy parameter is frozen, three descriptive audits are run on the anchor events; none carries an outcome, and none is a verdict on the decisions taken at the time. For Mati, with the first epoch at the documented detection time, exposure is reported

for the no-order policy, F1, F2, F2b, F3 and the oracle; the expectation stated beforehand, from the finding that Mati’s evacuation safety factor was well below one from detection [1], is that even the oracle’s exposure exceeds the noise band, and a disagreement is reported with its likely source. The smallest earlier detection at which the oracle’s exposure falls within the band is also reported ([R1-detection-grid]). For Varnavas, where the cell-broadcast message log can be assembled from public sources, the epoch at which F3 and the best fixed family would first have ordered each settlement is set beside the time of the actual message. Finally, the model’s clearance times for the Mati zones are set beside the evacuation times reported for Mati [8, 18], as a comparison, not a calibration target.

7. Results

7.1. Sequential rule against the best fixed trigger

Intent: Report the outcome of the sequential rule against the best training-tuned fixed trigger on held-out scenarios, with the misspecification and cost-ratio arms beside it.

7.2. Value-of-information ladder

Intent: Decompose the gain into updating the forecast, the change of rule form (reported only) and planning, and the perfect-information gap left.

7.3. Lead time against forecast skill

Intent: Report which lever lowers the cost more in fast and in slow scenarios, and how the advantage of lead over skill varies with the evacuation safety margin.

7.4. Forecast error and order breadth

Intent: Report whether the advantage of staged over simultaneous evacuation, at equal false alarms, changes when the forecast error is halved.

7.5. Staged against simultaneous evacuation (replication)

Intent: Report the staged-versus-simultaneous comparison at equal false alarms as a replication, with the descriptive non-monotonicity check.

7.6. Responder pre-positioning (secondary)

Intent: Report dynamic against static pre-positioning at equal unit-hours, with its dependence on the assumed responder effect stated.

7.7. Anchor events (descriptive)

Intent: Describe the anchor audits: consistency with published evacuation-safety findings for the first anchor, the order-time audit for the second, and the cross-model clearance check.

8. Discussion

Intent: Interpret the outcomes for evacuation practice: what forecast updating and planning are worth, when lead time beats forecast skill, and why broad early orders can congest the exits the threatened zones need.

9. Limitations

Intent: State that exposure is a proxy for harm, that results do not transfer to other regions or hazards without a new study, that historical decisions are not judged, and that the responder effect is an assumption.

10. Conclusion

Intent: Summarize the outcomes of the headline claims in ARGUMENT.md order and what a fire service can take from them.

Data and code availability

The simulation code, the scenario and experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The input data are public: road networks from OpenStreetMap, rapid-mapping products from the Copernicus Emergency Management Service, fire records from the European Forest Fire Information System, reanalysis weather from the Copernicus Climate Data Store and population grids from the Global Human Settlement Layer; access dates and snapshots are listed in the deposit.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

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During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the simulation and analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

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